

Department of Geology and Geophysics

Subject name: Electromagnetic methods of Prospecting (3-0-0) -3 Credit

Pre-requisite: None

Basic EM induction in the Earth, nature of propagation, sources of electromagnetic waves related to geophysics, mathematical representation of electromagnetic waves, Frequency spectrum and EM methods, EM reflection and transmissions of E and H fields, Basic concept and equations of electromagnetics, Maxwell's equations, boundary conditions, elliptic polarization, electromagnetic potential and waves, radiation from dipoles, retarded potential, near and far fields, radiation resistance, EM field of a loops of wire (electric and magnetic dipoles) on half-space, multi-layered media, impedance and its application. Diffusion equation and its solution, skin effect and skin depth. EM instrumentations, Various EM methods, classification on the basis of measured parameters, Dip angle, fixed transmitter inline and broad side arrays, Bieler Watson, Two-frame, Compansator and Turam method, Moving source and receiver (Slingram) method. AFMAG and VLF method, Airborne and transient EM methods, Type curves and phasor diagram in interpretation. EM scale modelling, Magnetotelluric methods, origin, theory and interpretation for 1-D, 2-D and 3-D. E- and H-polarizations, 3-D indicators. Marine EM surveys, Case studies related with various applications.

Text Books

1. Applied Geophysics, Telford et al, 1990
2. Geophysical Electromagnetic Theory and Methods, by MS Zhadnov, Elsevier
3. Practical Magnetotellurics, F Simpson and K Bar, Cambridge Uni. Press

Reference books

1. Potential Theory and Applied Geophysics, K.K. Roy
2. An Introduction to Geophysical Exploration, P. Keary

S.No	Topic	Number of lectures
1	Basic EM induction in the Earth, nature of propagation, sources of electromagnetic waves related to geophysics, mathematical representation of electromagnetic waves, Frequency spectrum and EM methods, EM reflection and transmissions of E and H fields,	3
2	Basic concept and equations of electromagnetics, Maxwell's equations, boundary conditions, elliptic polarization, electromagnetic potential and waves, radiation from dipoles, retarded potential, near and far fields, radiation resistance,	3
3	EM field of a loops of wire (electric and magnetic dipoles) on half-space, multi-layered media, impedance and its application. Diffusion equation and its solution, skin effect and skin depth. EM instrumentations,	4
4	Various EM methods, classification on the basis of measured parameters, Dip angle, fixed transmitter inline and broad side arrays, Bieler Watson, Two-frame, Compansator and Turam method,	5

5	Moving source and receiver (Slingram) method. AFMAG and VLF method,	4
6	Airborne and transient EM methods, Type curves and phasor diagram in interpretation. EM scale modelling,	4
7	Magnetotelluric methods, origin, theory and interpretation for 1-D, 2-D and 3-D. E- and H-polarizations, 3-D indicators.	5
8	Marine EM surveys, processing, interpretation	3
9	Case studies related with various applications	4
	Total	35